MA∗508 Time Series

Exam-like exercises for L9

Solve the following exercises.

1. Consider the causal AR(2) process $x_t = x_{t-1} - 0.5x_{t-2} + \epsilon_t$, where ϵ_t is white noise and $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$ of this AR(2) process.

2. Consider the causal AR(3) process $x_t = x_{t-1} - 0.4x_{t-2} + 0.3x_{t-3} + \epsilon_t$, where ϵ_t is white noise and $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$ of this AR(3) process.

3. Find the covariance function for a discrete-time weakly stationary process x_t with spectral density $f(\lambda) = 2 + \lambda^2$.